Intrinsic long range antiferromagnetic coupling in dilutely V doped $CuInTe_2$

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- **Collaborators**

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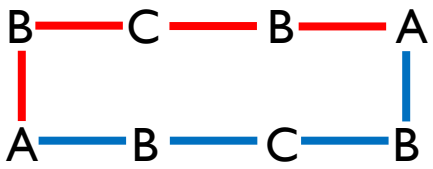
Xiaodong Zhang(postdoc), Bin Xi(undergrad), Junyi Zhu (advisor)

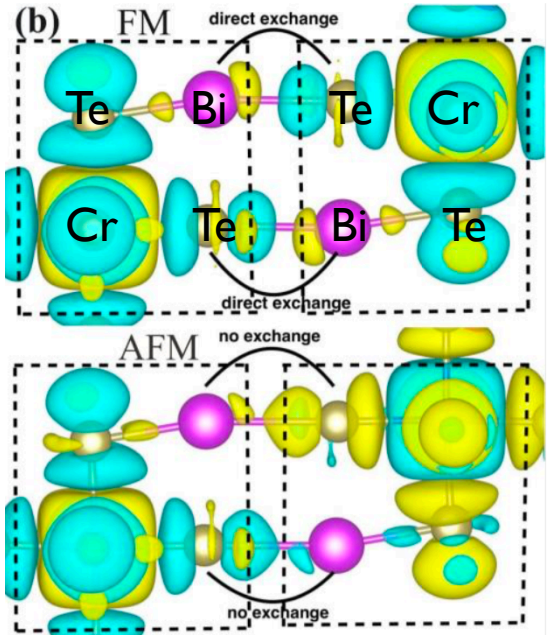
Background

- **Short range magnetic couplings:**
  - Double exchange (FM) (Zener, 1951),
  - p-d exchange (FM) (Zener, 1951),
  - Super-exchange (AFM & FM) (Anderson, 1950; Goodenough, 1955);
- **Long range magnetic couplings(with charge carriers) :**
  - RKKY interaction
- **Long range ferromagnetism(carrier free):**
  - Cr doped $(Bi, Sb)_2Te_3$ thin films (FM) (Chang, 2013)
- **Motivation:**
  - Search for carrier free long range AFM in diluted magnetic semiconductors
  - What is the theoretical model?

Phys. Rev. **81**, 440 (1951); Phys. Rev. **82,** 403 (1951);
Phys. Rev. **79,** 350 (1950); Phys. Rev. **100**, 564 (1955); Adv. Mater. **2013,** 25, 1065

Background





- **Previously we found that in Cr doped Bi_2Te_3**
  - **s lone pair states on C sites(stepping stone) mediate the long range FM order**
  - **Basic structure: ABCBA chain**
- **No theoretical model for this long range magnetic order**

Chin. Phys. Lett. **35** (1) 017502 (2018)

Hubbard model

- **Multi-bands Hubbard model**

$$H = \sum_{\langle i,j\rangle,\sigma} \sum_{\alpha\beta} t_{ij}^{\alpha\beta} c_{i\alpha\sigma}^{\dagger} c_{j\beta\sigma} + \sum_{i\alpha\sigma} \varepsilon_{i\alpha} n_{i\alpha\sigma} + \sum_i V_i$$

- **Local Coulomb interaction of p and d orbitals (Coury, 2016):**

$$V_p = \left(\frac{U}{2} - \frac{J}{4}\right) : n^2 : - \frac{J}{4} : \boldsymbol{m}^2 : + \frac{J}{2} \sum_{\alpha\beta} : n_{\alpha\beta}^2 : ,\ U = V_{\alpha\beta,\alpha\beta},\ J = V_{\alpha\beta,\beta\alpha}$$

$$V_d = \left(\frac{U}{2} - \frac{J}{4} + \frac{5}{2}\Delta J\right) : n^2 : - \left(\frac{J}{4} - \frac{3}{2}\Delta J\right) : \boldsymbol{m}^2 : + \left(\frac{J}{2} - 3\Delta J\right) \sum_{\alpha\beta} : n_{\alpha\beta}^2 : + \frac{2}{3}\Delta J : Q^2 :$$

$$U = V_{23,23},\ J = (V_{15,51} + V_{23,32})/2,\ \ \Delta J = V_{15,51} - V_{23,32}$$

- **We are going to use Coury's work as a starting point of modeling**

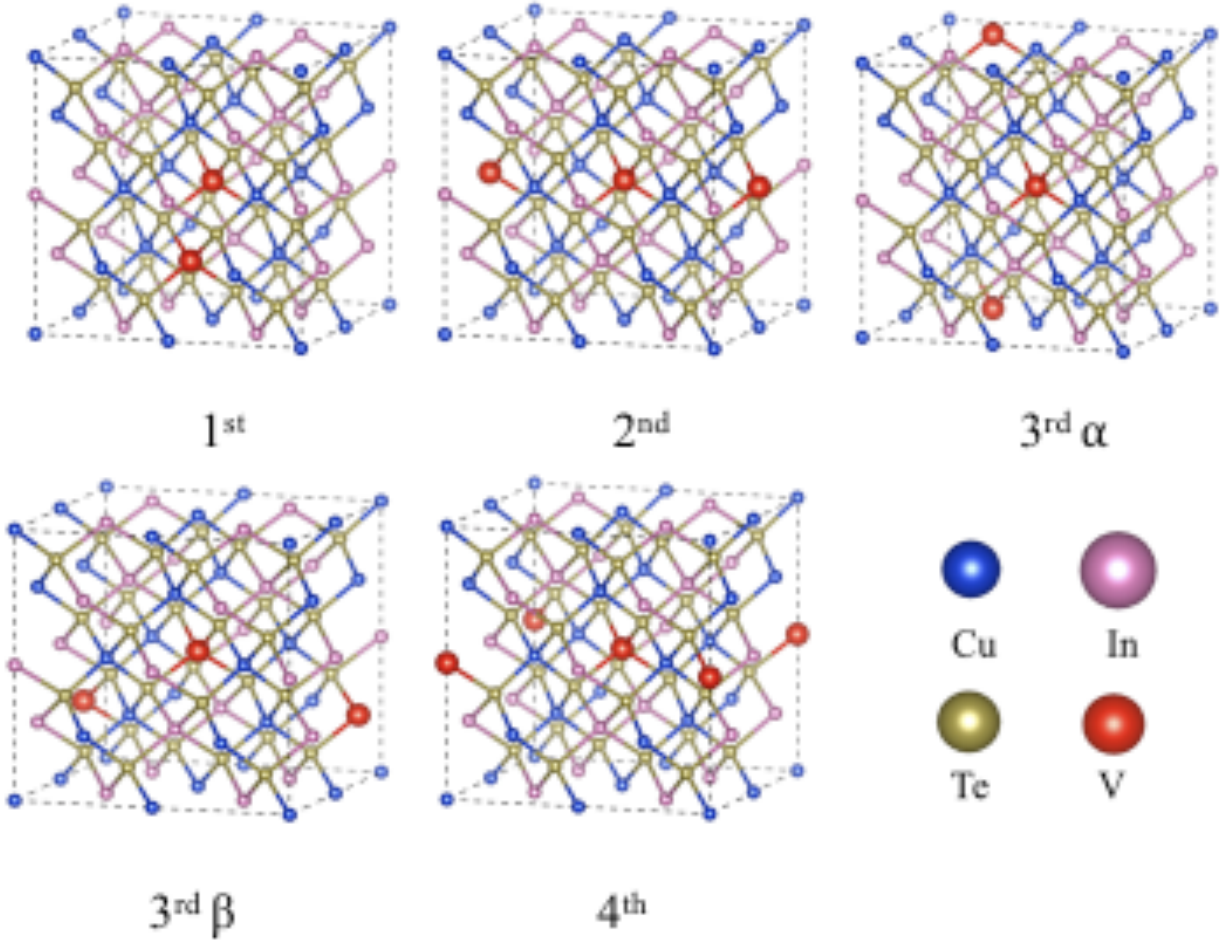
Phys. Rev. B **93**. 075101

Guidelines to search for long range magnetism in DMS

- Small doping concentration → **large distance between magnetic dopants**
- Nearest neighbor coupling strength$\sim t^2/\Delta E$ → **small energy difference**
- Coupling range longer than super-exchange(ABA) →**ABCBA**
- Stepping stone site has s lone pair states → **can it be d states?**
  - **More local orbital degeneracy**

DFT calculations settings

- **Calculation settings:**
  - GGA+U without/with SOC
  - 2x2x1& 3x3x1 supercell
    - Both show similar results
  - Convergence tests on k mesh and energy cut off were done
- **System:**
  - $CuInTe_2$, $CuGaTe_2$, $AgInTe_2$, $AgGaTe_2$
  - Use $CuInTe_2$ to demonstrate



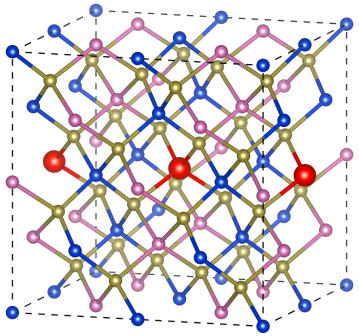
Configurations of $CuInTe_2$ 2x2x1 supercell

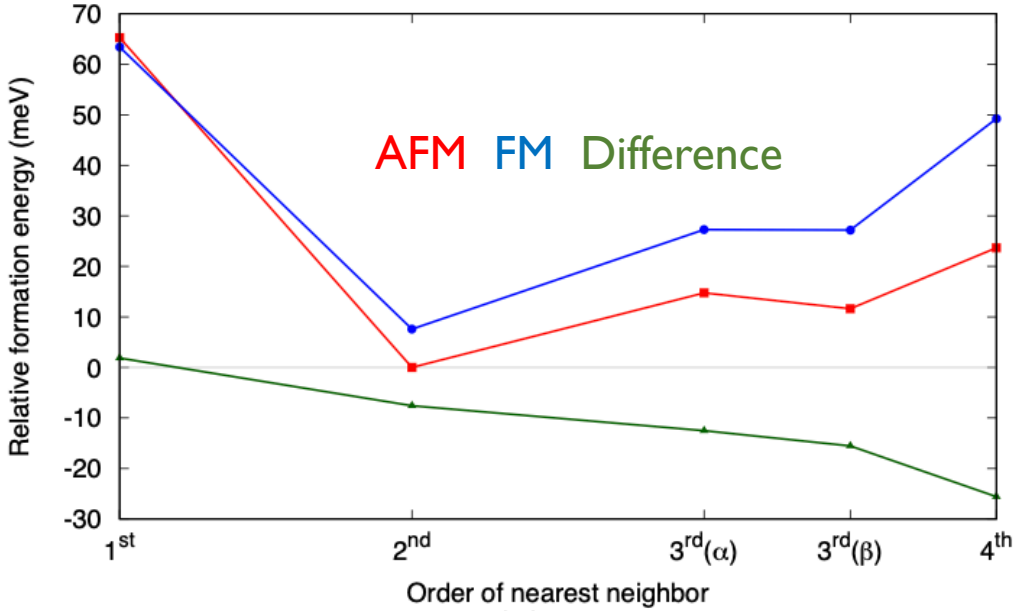
Relative formation energy in $CuInTe_2$

- **The relative formation energy is defined as:**

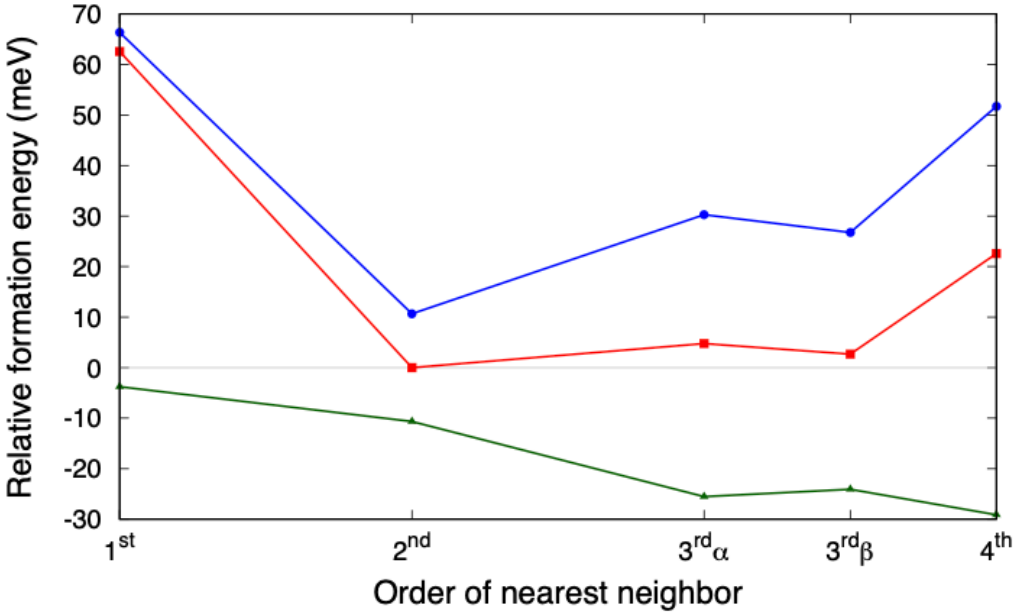
$$E_f = E_{doped} - E_{undoped} + n_{In}\mu_{In} - n_V\mu_V$$

- **2nd NN is the most stable one:**
  - AFM 7.6 meV (10.7 meV with SOC) lower than FM
  - V-Te-Cu(Stepping stone site)-Te-V chain structure
  - 6.3 Å between two V atoms





(a) no soc



(b) soc

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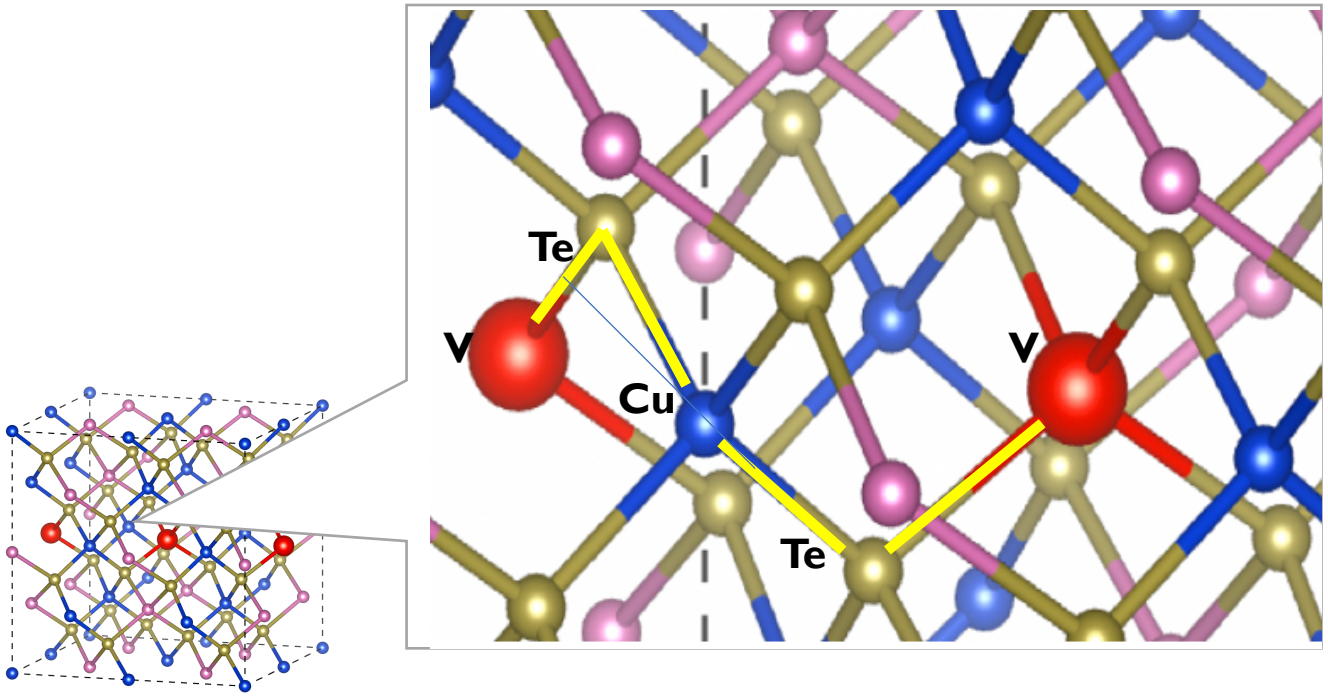
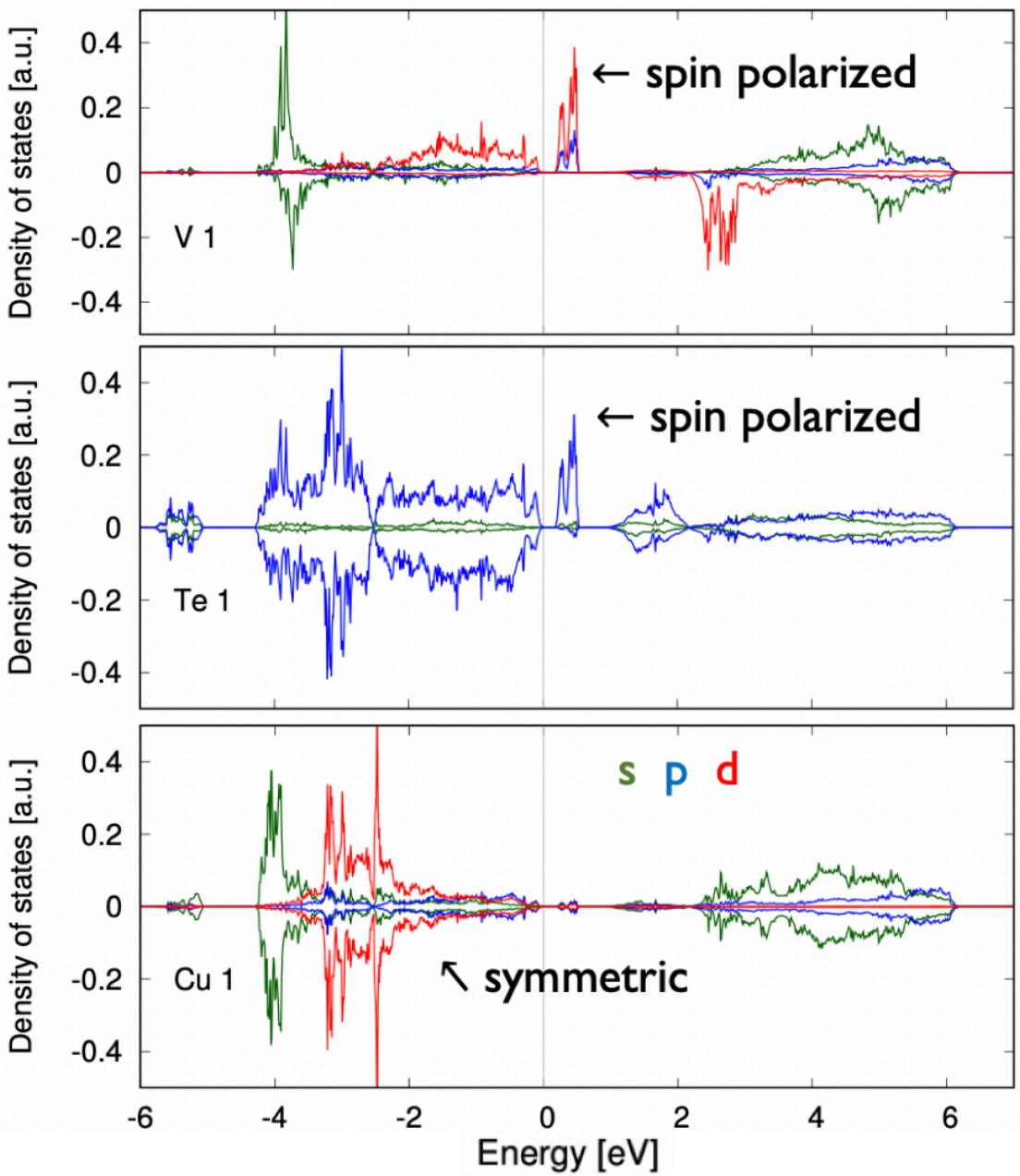
pDOS

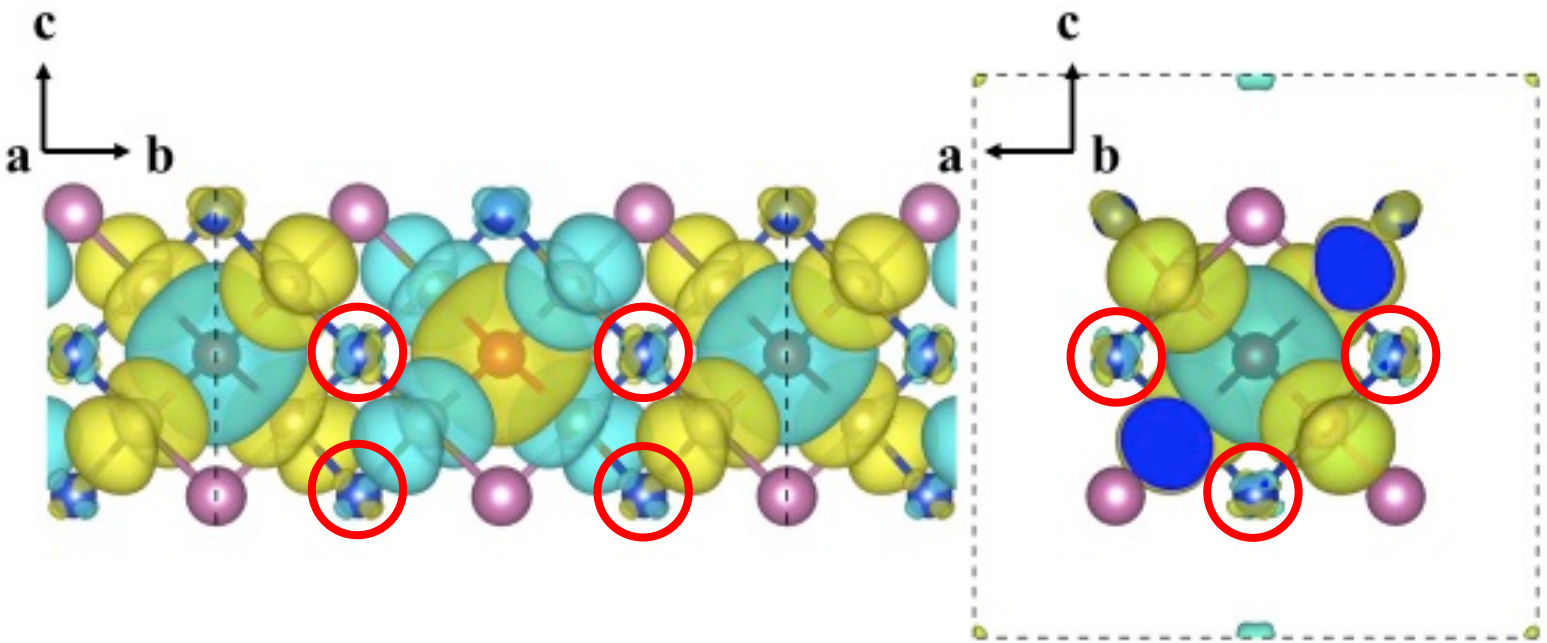
- Te on the chain are polarized
- The pDOS of Cu at stepping stone sites is symmetric
- Is it really unpolarized?



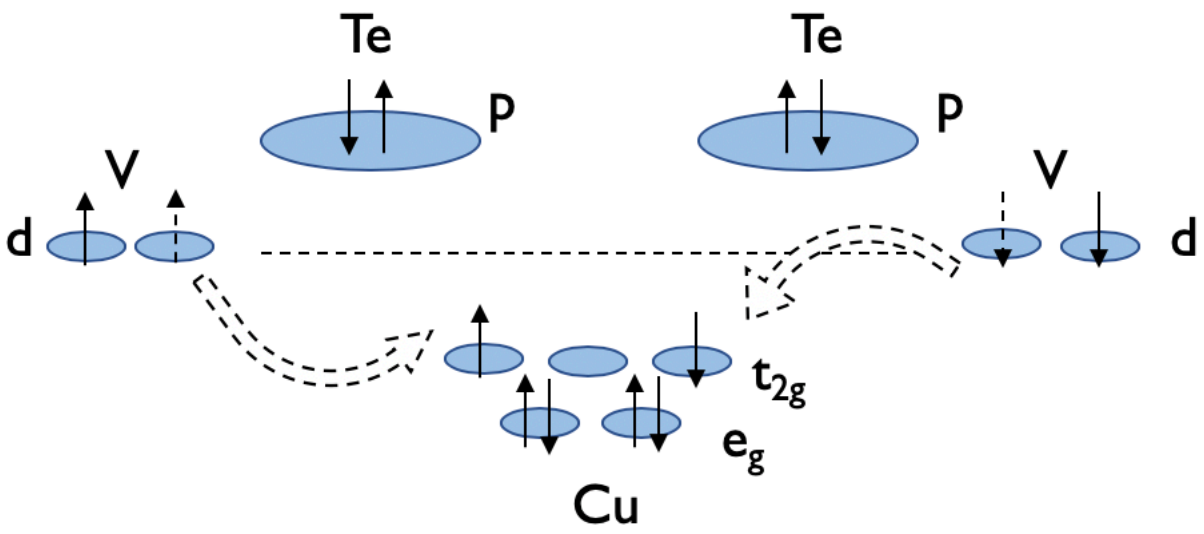


Spin texture

- **Polarized d states of Cu**
- **Verification:**
  - Replace Cu to Na
    7.57 meV → 1.6 meV



- **Serves as stepping stones to mediate long range magnetism**



Local Coulomb interaction terms

- **Using Coury's work as a starting point, we derived a multi-bands Hubbard model to explain the spin chain of ABCBA:**

$$H = \sum_{\langle i,j\rangle,\sigma} \sum_{\alpha\beta} t_{ij}^{\alpha\beta} c_{i\alpha\sigma}^{\dagger} c_{j\beta\sigma} + \sum_{i\alpha\sigma} \varepsilon_{i\alpha} n_{i\alpha\sigma} + \sum_{i} V_i \,, V_i = V_0^i + V_{sf}^i + V_{ph}^i$$

where

$$\hat{V}_0 = un^2 - vm_z^2 - (u-v)n + 8v \sum_{\alpha} n_{\alpha\uparrow} n_{\alpha\downarrow}$$

$$\hat{V}_{sf} = -2v \sum_{\alpha\neq\beta,\sigma} c_{\alpha\sigma}^{\dagger} c_{\alpha,-\sigma} c_{\beta,-\sigma}^{\dagger} c_{\beta\sigma}$$ spin flipping (local exchange)

$$\hat{V}_{ph} = -2v \sum_{\alpha\neq\beta,\sigma} c_{\alpha\sigma}^{\dagger} c_{\alpha,-\sigma}^{\dagger} c_{\beta\sigma} c_{\beta,-\sigma}$$ pair hopping

- **New terms coming from the local orbital degeneracy can give rise to richer phenomena than single band Hubbard model**
- **Derivation, different configurations, role of last 2 terms**

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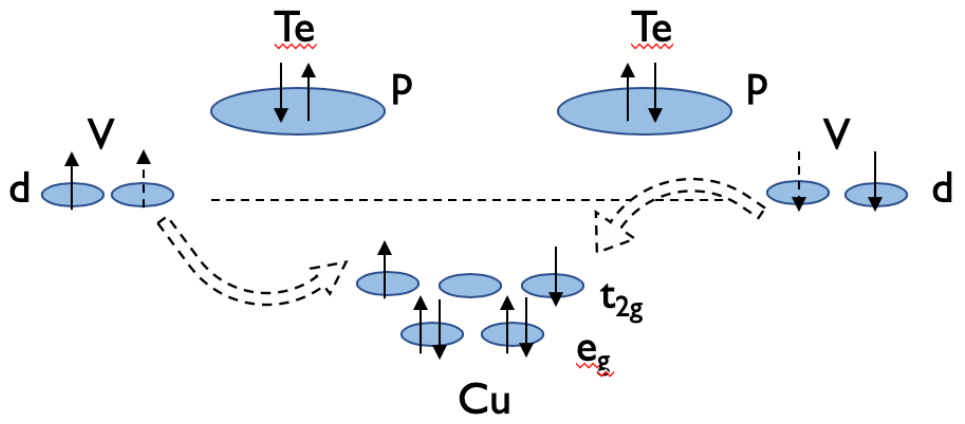
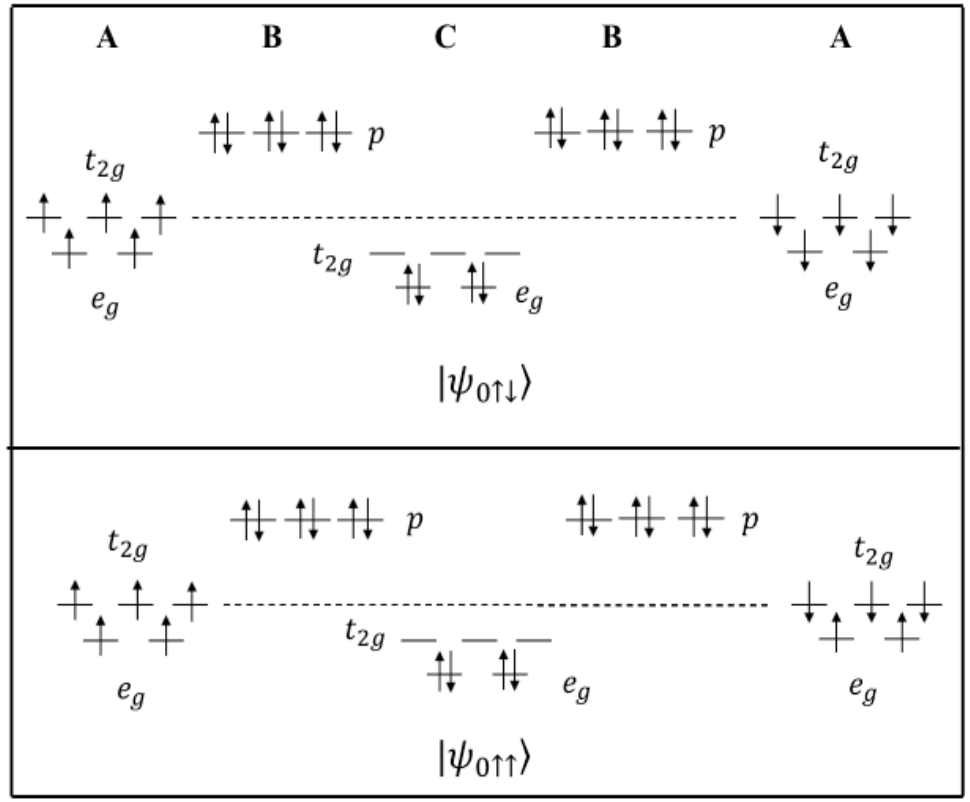
Apply to ABCBA Structure



- **Effective hopping between A and C**
- **An effective second order super-exchange: A-C-A**
- **The energy difference is**

$$E_{\uparrow\uparrow} - E_{\uparrow\downarrow} \approx 24v_A - \frac{72t^4 v_A}{\Delta E_{10}{}^2 \Delta E_{20}{}^2}$$

  - v_A is the parameter v on site A
  - ΔE_{10} and ΔE_{20} are the first and second excitation energy



- Verification:
  - Replace Cu to Ag: larger ΔE → larger magnetic coupling
  - DFT result: 7.57 meV → 8.89 meV

Conclusion

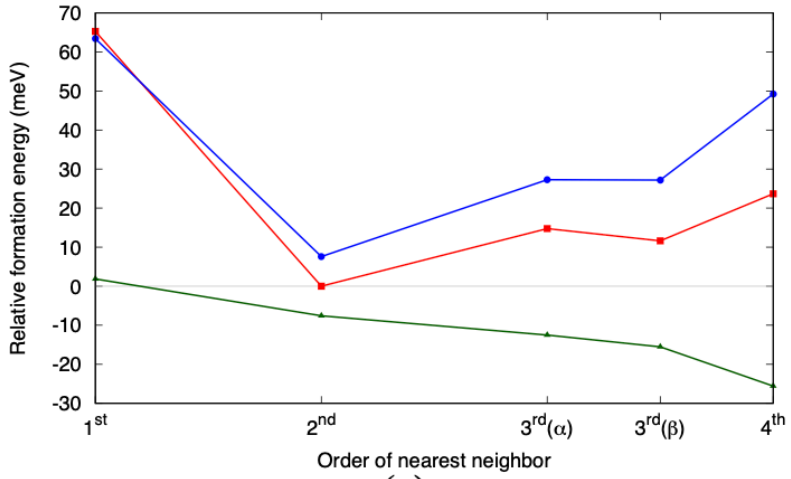
- **We found the first examples of intrinsic long range AFM coupling in $CuInTe_2$ and other chalcopyrite;**
- **We derived a multi-bands Hubbard model to give the mechanism**
- **This model can be applied to other DMS to find magnetic order**

Thank you

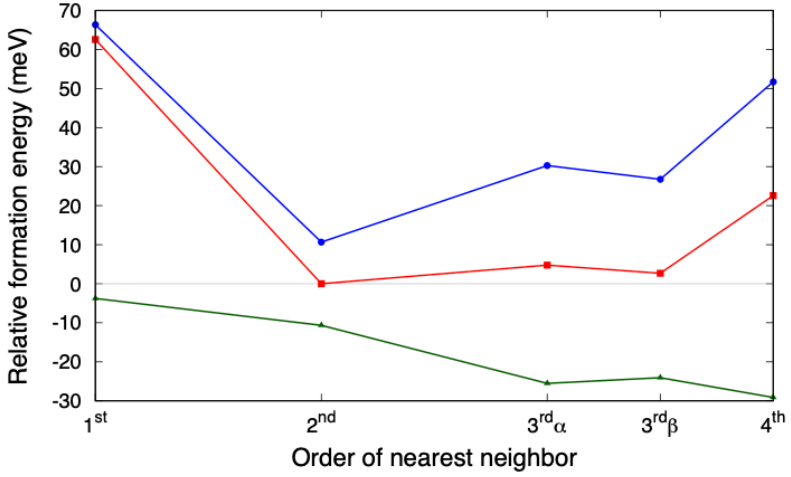
2x2x1 & 3x3x1supercell

- (a) no SOC (b) static SOC

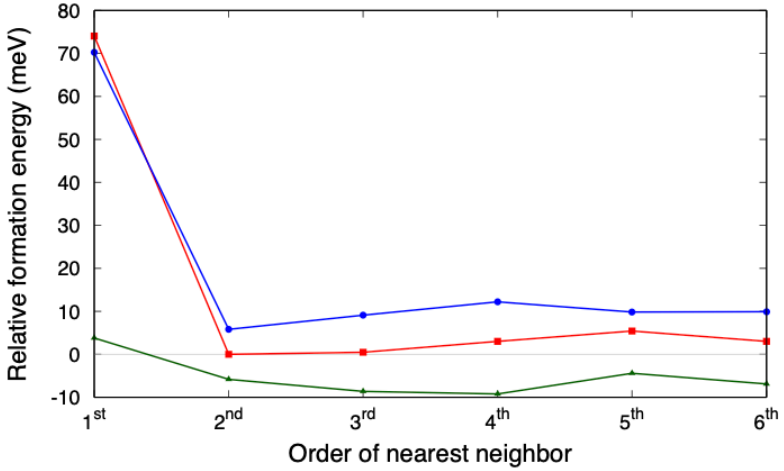
AFM FM Difference



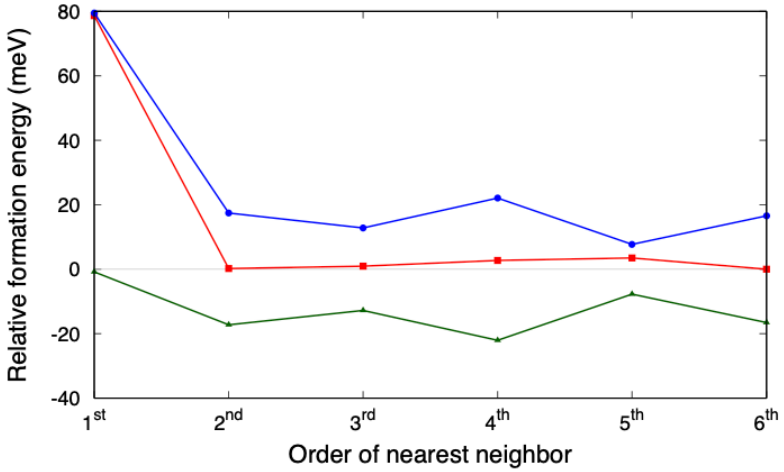
(a)



(b)



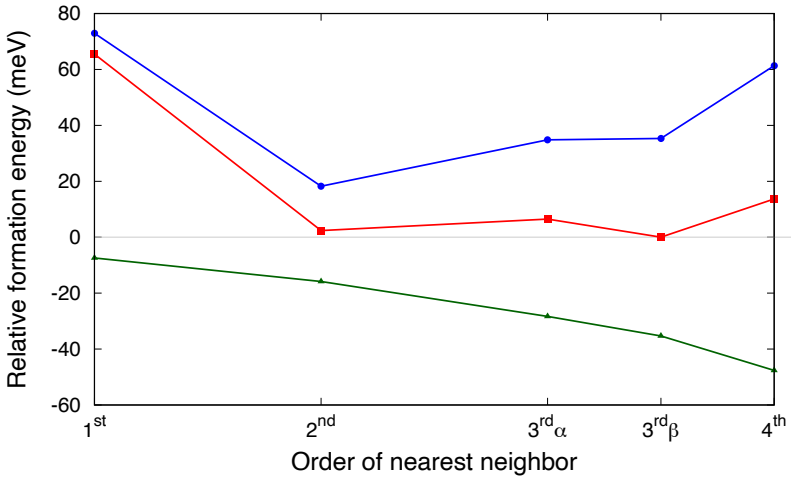
(a)

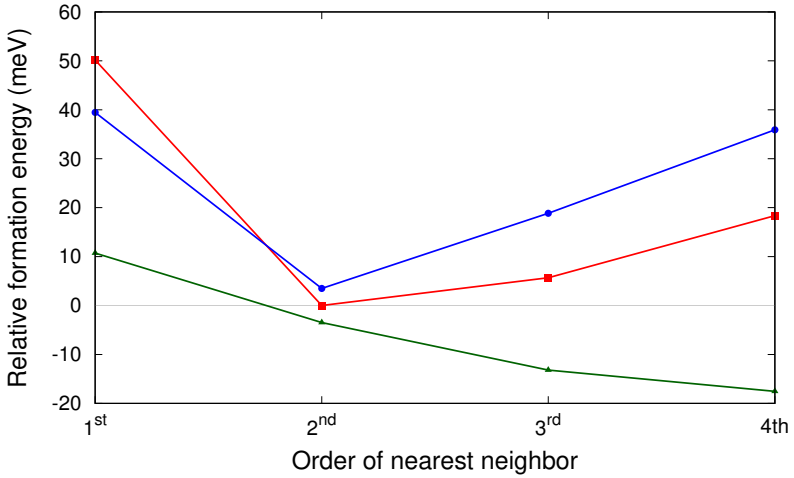


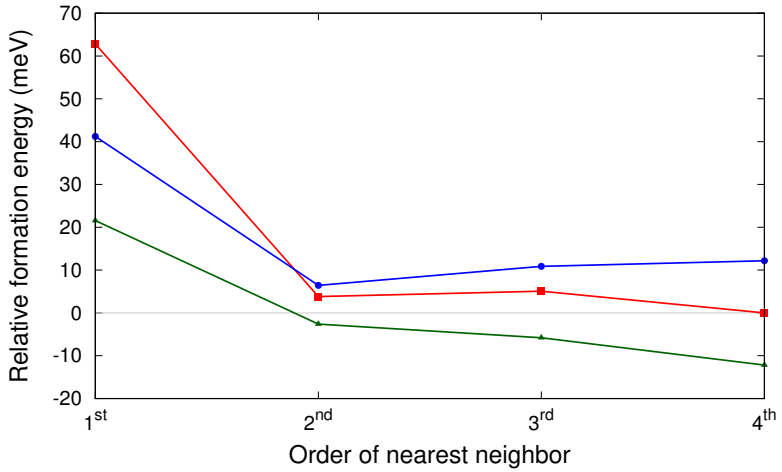
(b)

$CuGaTe_2$, $AgInTe_2$ & $AgGaTe_2$

- **Long range is still more stable**







AFM FM Difference